

The equations of the tangent and the normal

Lessons 17–18 · Algebra 11, §1.6 · P1 · 7.6 · 28 problems

Every answer needs one gradient and one point. The normal gradient is $-\frac{1}{m}$, so a horizontal tangent has a **vertical** normal, which has no gradient at all.

Easy 7

One derivative, one substitution, one line. Problem 7 is the trap: the tangent is horizontal, so the normal cannot be written as $y = mx + c$.

1 Tangent to $y = x^2 + 1$ at $x = 2$ $\left| y = 4x - 3 \right.$
 $y' = 2x = 4$ and the point is $(2, 5)$, so $y - 5 = 4(x - 2)$.

2 Normal to $y = x^2 + 1$ at $x = 2$ $\left| y = -\frac{x}{4} + \frac{11}{2} \right.$
 The tangent gradient is 4, so the normal gradient is $-\frac{1}{4}$ through $(2, 5)$.

3 Tangent to $y = x^3$ at $x = 1$ $\left| y = 3x - 2 \right.$
 $y' = 3x^2 = 3$ at the point $(1, 1)$.

4 Gradient of the normal to $y = x^2$ at $x = 3$ $\left| -\frac{1}{6} \right.$
 $y' = 6$, and the normal gradient is the negative reciprocal.

5 Where is the tangent to $y = x^2 + 8x$ horizontal? $\left| x = -4 \right.$
 $y' = 2x + 8 = 0$.

6 Tangent to $y = 5 - 2x$ at $x = 7$ $\left| y = 5 - 2x \right.$
 A straight line is its own tangent everywhere — the gradient never changes.

7 Normal to $y = x^3$ at $x = 0$ $\left| x = 0 \right.$
 $y'(0) = 0$, so the tangent is the x -axis and the normal is vertical. A vertical line has the equation $x = 0$, not $y =$ anything.

Medium 7

The derivative takes a line or two first. In 12 and 13 the gradient is given and the point must be found, which is the reverse of the usual order.

8 Tangent to $y = x^3 - 3x + 1$ at $x = 2$ $\left| y = 9x - 15 \right.$
 $y' = 3x^2 - 3 = 9$ and $y(2) = 3$, so $y - 3 = 9(x - 2)$.

9 Normal to $y = x^3 - 3x + 1$ at $x = 2$ $\left| y = -\frac{x}{9} + \frac{29}{9} \right.$
 Gradient $-\frac{1}{9}$ through $(2, 3)$. Multiplying through by 9 gives $9y + x = 29$.

10 Tangent to $y = \sqrt{x}$ at $x = 9$ $y = \frac{x}{6} + \frac{3}{2}$

$y' = \frac{1}{2\sqrt{x}} = \frac{1}{6}$ and the point is $(9, 3)$.

11 Tangent to $y = \frac{6}{x}$ at $x = 3$ $y = 4 - \frac{2x}{3}$

$y = 6x^{-1}$, so $y' = -6x^{-2} = -\frac{2}{3}$, and the point is $(3, 2)$.

12 Where has $y = x^2 - 2x$ a tangent parallel to $y = 4x$? $x = 3$

Parallel means equal gradients: $2x - 2 = 4$.

13 Where has $y = x^3 - 9x$ a tangent parallel to the x -axis? $x = \pm\sqrt{3}$

$3x^2 - 9 = 0$, so $x^2 = 3$. Two places, not one.

14 The tangent to $y = x^2$ at $x = 4$ crosses the x -axis where? $x = 2$

The tangent is $y = 8x - 16$; setting $y = 0$ gives $x = 2$. For $y = x^2$ the tangent always cuts the axis at half the x -coordinate.

Hard 7

The point of contact is unknown, so call it a and let the condition find it. In 16 the normal is intersected with the curve, which needs a quadratic, not a substitution.

15 The tangent to $y = x^2 - 3x$ at $x = a$ passes through $(0, -4)$. Find a . $a = \pm 2$

Gradient $2a - 3$ at $(a, a^2 - 3a)$, so the tangent is $y = (2a - 3)(x - a) + a^2 - 3a$. Putting $x = 0$ gives $-2a^2 + 3a + a^2 - 3a = -a^2$. Setting $-a^2 = -4$ gives $a^2 = 4$.

16 The normal to $y = x^2$ at $x = 2$ meets the curve again. Find the second point. $(-\frac{9}{4}, \frac{81}{16})$

The normal is $y = -\frac{x}{4} + \frac{9}{2}$. Solving $x^2 = -\frac{x}{4} + \frac{9}{2}$ gives $4x^2 + x - 18 = 0$, that is $(x - 2)(4x + 9) = 0$. The root $x = 2$ is the point we started from, so the new one is $x = -\frac{9}{4}$.

17 Find k so that $y = 2x + k$ is a tangent to $y = x^2 + 3x + 4$. $k = \frac{15}{4}$

A tangent touches, so the two meet in a repeated root. $x^2 + 3x + 4 = 2x + k$ gives $x^2 + x + (4 - k) = 0$, whose discriminant $1 - 4(4 - k)$ must be zero.

18 Find the tangent and the normal to $y = x^3 - x$ at $x = -1$. $y = 2x + 2; y = -\frac{x}{2} - \frac{1}{2}$

$y' = 3x^2 - 1 = 2$ and $y(-1) = 0$, so the point is $(-1, 0)$. The normal gradient is $-\frac{1}{2}$ through the same point.

- 19 Show that the tangent to $y = \sqrt{x}$ at $x = a$ cuts the y -axis at $(0, \frac{\sqrt{a}}{2})$. **shown**

$$y' = \frac{1}{2\sqrt{a}}, \text{ so the tangent is } y = \sqrt{a} + \frac{1}{2\sqrt{a}}(x - a). \text{ At } x = 0 \text{ this is } \sqrt{a} - \frac{a}{2\sqrt{a}} = \sqrt{a} - \frac{\sqrt{a}}{2} = \frac{\sqrt{a}}{2}.$$

- 20 Find the two tangents to $y = x^2$ that pass through $(1, -3)$. **$y = 6x - 9$ and $y = -2x - 1$**

The tangent at $x = a$ is $y = 2ax - a^2$. Through $(1, -3)$: $2a - a^2 = -3$, so $a^2 - 2a - 3 = 0$ and $a = 3$ or $a = -1$. The point lies below the parabola, which is why there are two.

- 21 Where do the normals to $y = x^2$ at $x = 1$ and $x = -1$ meet? **$(0, \frac{3}{2})$**

At $x = 1$: gradient $-\frac{1}{2}$ through $(1, 1)$, so $y = -\frac{x}{2} + \frac{3}{2}$. At $x = -1$: gradient $\frac{1}{2}$ through $(-1, 1)$, so $y = \frac{x}{2} + \frac{3}{2}$. By symmetry they meet on the y -axis.

Very hard 7

General results, and one that is easier by coordinate geometry than by calculus. In 25 the algebra is the proof: the second root is forced to exist.

- 22 The tangent to $y = \frac{1}{x}$ at $x = a$ ($a > 0$) meets the axes at P and Q . Show that the point of contact is the midpoint of PQ . **shown**

$y' = -\frac{1}{a^2}$, so the tangent is $y = \frac{2}{a} - \frac{x}{a^2}$. It meets the x -axis at $P(2a, 0)$ and the y -axis at $Q(0, \frac{2}{a})$. The midpoint of PQ is $(a, \frac{1}{a})$, which is the point of contact. True for every a , which is why the triangle it cuts has constant area.

- 23 Find the angle between $y = x^2$ and $y = x^3$ at $(1, 1)$. **$\approx 8.13^\circ$**

The gradients there are 2 and 3. For two lines, $\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right| = \frac{1}{7}$, so $\theta = \arctan \frac{1}{7} \approx 8.13^\circ$. The curves are nearly parallel at that point.

- 24 Find the normal to $y = x^2 - 4x$ at the point where the curve crosses the positive x -axis. **$y = 1 - \frac{x}{4}$**

$x^2 - 4x = 0$ gives $x = 0$ or $x = 4$, and the positive one is $x = 4$. $y' = 2x - 4 = 4$, so the normal gradient is $-\frac{1}{4}$ through $(4, 0)$.

- 25 Show that every normal to $y = x^2$ except the y -axis meets the curve twice. **shown**

At $x = a$ with $a \neq 0$ the normal is $y = a^2 - \frac{1}{2a}(x - a)$. Solving against $y = x^2$ gives a quadratic whose roots are $x = a$ and $x = -a - \frac{1}{2a}$. These agree only if $2a^2 + 1 = -2a^2$, that is $4a^2 = -1$, which is impossible — so the two points are always distinct. At $a = 0$ the normal is the y -axis, which meets the parabola only at the origin.

- 26 Find the points on $y = x^3 - 3x^2 + 2$ where the tangent is parallel to $y = 9x$. **$(3, 2)$ and $(-1, -2)$**

$y' = 3x^2 - 6x = 9$ gives $x^2 - 2x - 3 = 0$, so $x = 3$ or $x = -1$. Substituting back into the curve, not into the derivative, gives the y -coordinates.

27 The tangent to $y = x^2 + bx + c$ at $x = 1$ is $y = 3x - 2$. Find b and c . $b = 1, c = -1$

Two conditions. Equal gradients: $y'(1) = 2 + b = 3$, so $b = 1$. Same point: the line gives $y = 1$ at $x = 1$, and the curve gives $1 + b + c$, so $2 + c = 1$. Check: $y = x^2 + x - 1$ has $y(1) = 1$ and $y'(1) = 3$. ✓

28 Find the shortest distance from the origin to the tangent to $y = x^2$ at $x = 1$. $\frac{\sqrt{5}}{5} \approx 0.447$

The tangent is $y = 2x - 1$, or $2x - y - 1 = 0$. The distance from $(0, 0)$ to $Ax + By + C = 0$ is $\frac{|C|}{\sqrt{A^2 + B^2}} = \frac{1}{\sqrt{5}}$. Calculus would work too, but it is a longer road to the same number.
